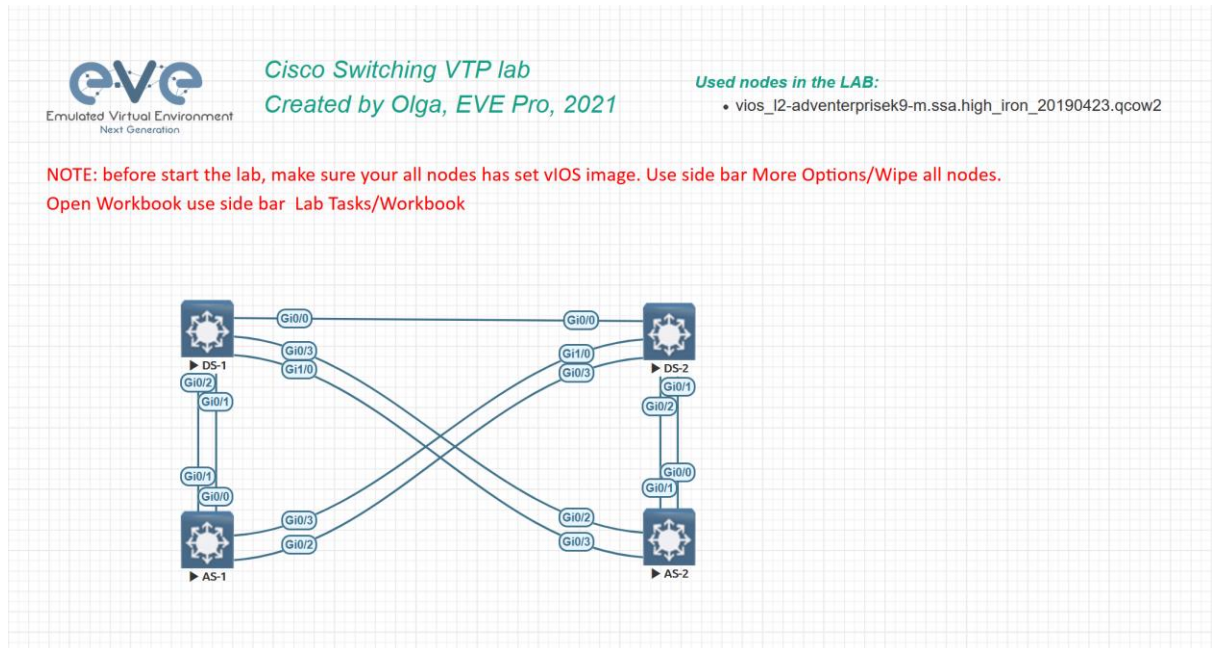


CCNP Switching VTP Lab

Topology



Objectives

1. Configure Basic Device Settings
2. Implement and Observe a VTPv2 Domain
3. Implement and Observe a VTPv3 Domain

VLAN Trunking Protocol (VTP) is a technology that manages the addition, deletion, and renaming of VLANs on the entire network from a single switch. Although not commonly used in modern networks that support programmability, VTP can be a useful tool as long as it is carefully managed.

A VTP domain, also called a *VLAN management domain*, consists of trunked switches that are under the same administrative control sharing the same VTP domain name. A switch can be in only one VTP domain, and VLAN database contents in the domain are globally synchronized. VLAN information is not propagated until a domain name is specified and trunks are set up between the devices.

There are three versions of VTP available; version 1 is the default. By default, versions 1 and 2 can support only normal-range VLANs. Version 3 can support normal and extended-range VLANs, as well as the synchronization of other databases, like MST. Support for VTP Version 3 on the Catalyst platform was added in IOS version 12.2(52)SE. Older IOS versions do not generally support VTP Version 3.

Switches operate in one of four VTP modes. The default VTP mode is server mode.

VTP Mode	Description
VTP Server	You can create, modify, and delete VLANs, as well as specify other configuration parameters, such as VTP version and VTP pruning, for the entire VTP domain. VTP servers advertise their VLAN configuration to other switches in the same VTP domain and synchronize their VLAN configuration with other switches based on advertisements received over trunk links. VTP server is the default mode. In VTP Server mode, VLAN configurations are only stored in the flash:vlan.dat file. While VLANs are manipulated in the configuration mode, the configuration commands do not appear in the running-config.
VTP Client	A VTP client behaves like a VTP server and transmits and receives VTP updates on its trunks, but you cannot create, change, or delete VLANs on a VTP client. VLANs are configured on another switch in the domain that is in server mode. In VTP Client mode, learned VLANs are only stored in the flash:vlan.dat file. The configuration of VLANs does not appear in the running-config.
VTP Transparent	VTP transparent switches do not participate in VTP. A VTP transparent switch does not advertise, nor synchronize, its VLAN database based on received advertisements. However, transparent switches will forward received VTP messages under two circumstances; either the VTP domain name of the transparent switch is empty (not yet configured), or it matches the domain name in the received VTP messages. In VTP Transparent mode, normal range VLAN configurations are stored both in flash:vlan.dat file and they are also present in the running-config. If extended range VLANs are used, they are stored in the flash:vlan.dat only if the switch is running VTP version 3. In Version 1 or 2, extended VLANs are stored in the running-config.
VTP Off	A switch in VTP Off mode functions in the same manner as a VTP transparent switch, except that it does not forward VTP advertisements on trunks. VTP off is only available on switches that support VTP version 3, although it is not necessary to run VTP version 3 on the switch to be able to put it into VTP Off mode. In VTP Off mode, normal range VLAN configurations are stored both in flash:vlan.dat file and are also present in the running-config. Similar to Transparent mode, when using VTP version 3 both normal and extended range VLANs are stored in the flash:vlan.dat file.

In this lab you will set up and observe the operation of VTP version 2 and VTP version 3.

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Task 1: Configure Basic

Switch DS-1

```
hostname DS-1
spanning-tree mode rapid-pvst
line con 0
exec-timeout 0 0
logging synchronous
exit
interface range g0/0-3, g1/0-3
shutdown
exit
interface range g0/0-3, g1/0
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
exit
```

Switch DS-2

```
hostname DS-2
spanning-tree mode rapid-pvst
line con 0
exec-timeout 0 0
logging synchronous
exit
interface range g0/0-3, g1/0-3
shutdown
exit
interface range g0/0-3, g1/0
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
exit
```

Switch AS-1

```
hostname AS-1
spanning-tree mode rapid-pvst
line con 0
exec-timeout 0 0
logging synchronous
exit
interface range g0/0-3, g1/0-3
shutdown
exit
interface range g0/0-3
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
exit
```

Switch AS-2

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```
hostname AS-2
spanning-tree mode rapid-pvst
line con 0
  exec-timeout 0 0
  logging synchronous
  exit
interface range g0/0-3, g1/0-3
  shutdown
  exit
interface range g0/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  no shutdown
  exit
```

Task 2: Implement and Observe a VTPv2 Domain

Step 1. *Verify VTP status*

On DS-1, issue the command **show vtp status**.

```
DS-1#sh vtp status
VTP Version capable           : 1 to 3
VTP version running           : 1
VTP Domain Name                :
VTP Pruning Mode               : Disabled
VTP Traps Generation           : Disabled
Device ID                     : 500a.0001.8000
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN:
-----
VTP Operating Mode             : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs       : 5
Configuration Revision          : 0
MD5 digest                     : 0x57 0xCD 0x40 0x65 0x63 0x59 0x47 0xBD
                               : 0x56 0x9D 0x4A 0x3E 0xA5 0x69 0x35 0xBC
```

Because no VLAN configurations were made, all settings are the defaults. This switch is capable of running versions 1, 2, or 3 of VTP, and runs version 1 by default. All switches in the VTP domain must run the same VTP version. The number of existing VLANs is the five built-in VLANs. Different switches in the Catalyst family support different numbers of local VLANs. Lastly, note that the configuration revision is 0.

Multiple switches in the VTP domain can be in VTP Server mode. In VTPv1 and VTPv2, any of these server switches can be used to centrally manage VLANs for all other switches in the VTP domain. The configuration revision number is compared amongst VTPv1 or VTPv2 switches. The VLAN database from the switch with the highest revision number is adopted by all the other switches in the VLAN management domain. Every time VLAN information is modified and saved in the VLAN database (vlan.dat), the revision number is increased by one when the user exits from VLAN Configuration mode.

VTP messages only pass over trunks between switches. They are not sent out of access ports.

Task 3: Configure and observe VTPv2 domain operations.

Step 1. Configure VTPv2

Configure DS-1 to operate in VTP Server mode and set the VTP domain name and VTP version 2. We will also set a VTP password, which provides some rudimentary protection against automatic VLAN database propagation. Because this password is set, VTPv2 will not allow other switches to automatically learn the domain name.

```
DS-1# config t
Enter configuration commands, one per line. End with CNTL/Z.
DS-1(config)# vtp domain CCNPv8
Changing VTP domain name from NULL to CCNPv8
DS-1(config)# vtp version 2
DS-1(config)# vtp mode server
Device mode already VTP Server for VLANs.
DS-1(config)# vtp password cisco123
Setting device VTP password to cisco123
DS-1(config)#
Jan 6 21:39:03.990: %SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name
changed to CCNPv8.
```

On DS-1, create VLAN 2 and name it SecondVLAN, assign port ~~g1/0/23~~ ^{Gi1/1} to that VLAN, and verify the VTP status again. Notice the revision number has incremented. Also, verify that the port is assigned to the VLAN.

```
DS-1# config t
Enter configuration commands, one per line. End with CNTL/Z.
DS-1(config)# vlan 2
DS-1(config-vlan)# name SecondVLAN
DS-1(config-vlan)# exit
DS-1(config)# interface g1/1
DS-1(config-if)# switchport mode access
DS-1(config-if)# switchport access vlan 2
DS-1(config-if)# no shutdown
DS-1(config-if)# end
```

```
DS-1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi1/2, Gi1/3
2	SecondVLAN	active	Gi1/1
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

```
DS-1#show vtp status
```

```
VTP Version capable      : 1 to 3
VTP version running      : 2
VTP Domain Name          : CCNPv8
VTP Pruning Mode         : Disabled
```

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```
VTP Traps Generation          : Disabled
Device ID                    : 500a.0001.8000
Configuration last modified by 0.0.0.0 at 1-10-21 21:01:46
Local updater ID is 0.0.0.0 (no valid interface found)
```

Feature VLAN:

```
VTP Operating Mode           : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs      : 6
Configuration Revision        : 2
MD5 digest                   : 0x36 0x8B 0x44 0xC9 0x2A 0x7E 0x1E 0x36
                              0x88 0x18 0x43 0xB2 0xF5 0xEC 0x2F 0x12
```

On D1, create VLAN 2048.

Note: If you are working this lab on something other than an IOS XE based switch, you will not be allowed to create VLAN 2048 in VTP version 1 or 2 in Server mode.

```
DS-1# config t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
DS-1(config)# vlan 2048
```

```
DS-1(config-vlan)# exit
```

Verify DS-2, AS-1 and AS-2 have not learned about the new VLANs by examining the output of **show vlan brief** on each switch.

Configure DS-2, AS-1 and AS-2 with the same VTP domain name, version and password as configured on DS-1.

Instead of waiting for the DTP update timer to expire, shut down and enable the interface g0/0 on DS-2 and f0/1 on A1 to force a DTP update, then verify that D2 and A1 now have the same VTP configuration revision number. However, they do not have the same VLAN database. D1 does not propagate information about extended VLANs (VLAN 2048) via VTP version 2. Notice from the output of **show vlan brief** that the only thing learned was the VLAN number and name; the port assignment was not propagated. Port assignments to VLANs still have to be done on per-switch basis.

```
DS-2# show vtp status | i VLANs|Revision
```

```
Maximum VLANs supported locally : 1005
```

```
Number of existing VLANs       : 6
```

```
Configuration Revision         : 2
```

```
DS-2# show vlan brief
```

```
DS-2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi1/1, Gi1/2, Gi1/3
2	VLAN0002	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

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To demonstrate the main drawback to VTP, go to AS-1 and remove VLAN 2 with the **no vlan 2** command. The command should complete successfully.

```
AS-1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
AS-1(config)# no vlan 2
AS-1(config)# end
```

```
AS-1# show vtp status | i Revision
```

```
Configuration Revision : 3
```

Go to DS-1 and see if VLAN 2 still exists. You will see that it does not. This is the main drawback to VTP version 2. Without careful control, an existing switch or newly added switch with the same VTP domain name information could overwrite the existing database, because it only compares the configuration revision number, instead of the Client / Server mode. Also, note the status of interface g1/1. It is not listed in the output of **show vlan brief**. The interface is unusable, because it is assigned to a VLAN that does not exist. Remember that the VLAN database contents and the port assignment to a VLAN are two separate things, and they are not synchronized.

```
D1# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi1/2, Gi1/3
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	
2048	VLAN2048	active	

```
DS-1# show vtp status
```

```
VTP Version capable      : 1 to 3
VTP version running      : 2
VTP Domain Name          : CCNPv8
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : d8b1.9028.af80
Configuration last modified by 0.0.0.0 at 1-6-20 21:57:29
Local updater ID is 0.0.0.0 (no valid interface found)
```

```
Feature VLAN:
```

```
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 5
Configuration Revision   : 3
MD5 digest               : 0xBE 0xF6 0xBD 0x14 0xED 0xA1 0x19 0x6A
                        : 0x3D 0x1C 0x22 0xB5 0x5E 0xC5 0x2C 0xE3
```

On DS-1, recreate VLAN 2 with the name SecondVLAN2. Verify that the VTP configuration revision number increases, the VLAN appears in the database, and that g1/0/23 is listed in the **show vlan brief** output once again.

```
DS-1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
DS-1(config)# vlan 2
DS-1(config-vlan)# name SecondVLAN2
```

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```
DS-1(config-vlan)# end
```

```
DS-1# show vtp status | i Revision
```

```
Configuration Revision : 4
```

```
DS-1# show vlan brief | i Gi1/1
```

```
2    SecondVLAN2          active    Gi1/1
```

Step 2. Configure AS-1 in VTP client mode

On AS-1, change the VTP mode to client, and then attempt to create VLAN 3. It will not be allowed. VTP Client mode prevents new VLANs from being created, modified, or deleted. But a VTP Client with a higher revision number will still overwrite a VTP Server switch database. The typical scenario is a switch using the same VTP information (domain, password) in a test network has a higher revision number and is placed on the production network without being cleared.

```
AS-1# config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
AS-1(config)# vtp mode client
```

```
Setting device to VTP Client mode for VLANs.
```

```
AS-1(config)# vlan 3
```

```
VTP VLAN configuration not allowed when device is in CLIENT mode.
```

Step 3. Configure DS-2 in VTP transparent mode

On DS-2, change the VTP mode to transparent, then attempt to create VLAN 3 with the name ThirdVLAN. You will succeed. Verify that it is in the output of **show vlan brief**. Note the VTP configuration revision number.

```
DS-2# config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
DS-2(config)# vtp mode transparent
```

```
Setting device to VTP Transparent mode for VLANs.
```

```
DS-2(config)# vlan 3
```

```
DS-2(config-vlan)# name ThirdVLAN
```

```
DS-2(config-vlan)# end
```

```
DS-2# show vlan brief | i ThirdVLAN
```

```
3    ThirdVLAN            active
```

```
DS-2# show vtp status | i Domain|Revision
```

```
VTP Domain Name : CCNPv8
```

```
Configuration Revision : 0
```

On DS-2, create VLAN 1111 with the name QuadOne. You will succeed. Verify that it is in the output of **show vlan brief**. Check to see what the VTP configuration revision number is.

```
DS-2# config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
DS-2(config)# vlan 1111
```

```
DS-2(config-vlan)# name QuadOne
```

```
DS-2(config-vlan)# end
```

```
DS-2# show vlan brief | i QuadOne
```

```
1111 QuadOne             active
```



```
DS-2# show vtp status | i Domain|Revision
VTP Domain Name      : CCNPv8
Configuration Revision : 0
```

Check DS-1 to see if VLAN 3 or VLAN 1111 has been learned. They will not have been. VTP Transparent switches keep to themselves in the VTP domain, and do not learn or propagate VLAN information from other switches.

On DS-1, shutdown interfaces g0/1 and g0/2, then create VLAN 4 with the name FourthVLAN. Verify that it is in the database, and that the VTP configuration revision number has incremented.

```
D1# show vtp status | i Revision
Configuration Revision : 4
```

```
DS-1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
DS-1(config)# interface range g0/1-2
DS-1(config-if-range)# shutdown
DS-1(config-if-range)# exit
DS-1(config)#
DS-1(config)# vlan 4
DS-1(config-vlan)# name FourthVLAN
DS-1(config-vlan)# end
```

```
DS-1# show vlan brief | i FourthVLAN
4      FourthVLAN      active
```

```
DS-1# show vtp status | i Revision
Configuration Revision : 5
```

Check DS-2 and AS-1 to see if they have learned about the existence of VLAN 4. DS-1 sent a VTP message to DS-2, which forwarded the VTP message to AS-1 because they shared the same VTP domain name. AS-1 added the VLAN to its database and incremented its CR number to 5. Because DS-2 is in Transparent mode, it does not add the VLAN to its database.

```
DS-2# show vtp status | i Revision
Configuration Revision : 0
```

```
AS-1# show vtp status | i Revision
Configuration Revision : 5
```

```
AS-1#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gil/0, Gil/1, Gil/2, Gil/3
2	SecondVLAN2	active	
4	FourthVLAN	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

Task 4: Implement and Observe a VTPv3 Domain

In this part of the lab you will configure VTP version 3 to operate across the rest of the switched network. VTP version 3 provides some significant benefits to the network administrator:

- VTP version 3 has the ability to hide the VTP password. On a VTP version 1 or 2 switch, issuing the command **show vtp password** will show the password to you in plaintext. VTP version 3 allows you to specify that the password be hidden in the output, preventing the password from being inadvertently or maliciously divulged.
- VTP version 3 can propagate information about extended range VLANs; These are VLANs numbered between 1006 and 4094. To support these VLANs with VTP version 1 or 2, all switches have to be in Transparent or Off mode and the VLANs must be configured manually on a switch-by-switch basis.
- VTP version 3 only supports pruning for normal-range VLANs.
- VTP version 3 supports propagating Private VLAN information. As with extended-range VLANs, the lack of PVLAN support in VTP version 2 required that all switches be in Transparent mode and manually configured at each switch.
- VTP version 3 added support for opaque databases. In other words, VTP version 3 can transport more than just the VLAN database between switches. The only option at this time is to share the Multiple Spanning Tree (MSTP) database, but room was left for expansion.
- Regardless of the VTP operating mode, a VTP domain must first be set before configuring VTP version 3.

VTP version 3 is backwards compatible with VTP version 2 for normal range VLANs only; at the boundary of the two protocols, a VTP version 3 switch will send out both version 3 and version 2-compatible messages. Version 2 messages received by a version 3 switch are discarded.

Step 1. Configure and verify VTPv3 on DS-1.

On DS-1, change the VTP version to version 3 and verify the change.

```
DS1# config t
Enter configuration commands, one per line. End with CNTL/Z.
DS-1(config)# vtp version 3
D1(config)#
*Jan 12 10:35:30.044: %SW_VLAN-6-OLD_CONFIG_FILE_READ: Old version 2 VLAN
confi
guration file detected and read OK. Version 3
files will be written in the future.
DS-1(config)# end

DS-1# show vtp status
VTP Version capable          : 1 to 3
VTP version running         : 3
VTP Domain Name              : CCNPv8
VTP Pruning Mode             : Disabled
VTP Traps Generation         : Disabled
Device ID                    : d8b1.9028.af80

Feature VLAN:
-----
VTP Operating Mode          : Server
Number of existing VLANs    : 7
Number of existing extended VLANs : 1
Maximum VLANs supported locally : 4096
Configuration Revision       : 0
```

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```
Primary ID           : 0000.0000.0000
Primary Description   :
MD5 digest           :
```

Feature MST:

```
VTP Operating Mode      : Transparent
```

DS-1#show vlan brief

VLAN Name	Status	Ports
1 default	active	Gi1/2, Gi1/3
2 SecondVLAN2	active	Gi1/1
4 FourthVLAN	active	
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	
2048 VLAN2048	active	

On DS-1, try to create VLAN 6. You will not be allowed to do so. If you examine the output of **show vtp status**, you will see that the identification of the primary server is blank (all zeroes).

DS-1# **show vtp status | i Primary**

```
Primary ID           : 0000.0000.0000
Primary Description   :
```

On DS-1, issue the privileged EXEC command **vtp primary vlan**. This makes DS-1 the primary for the VLAN database only. There is also a primary for MST, but that is not the focus of this lab (if you issue the **vtp primary** command with no additional parameters, VLAN is assumed.) Verify that the base mac-address for DS-1 is listed as the primary server id.

DS-1# **vtp primary vlan**

This system is becoming primary server for feature vlan

No conflicting VTP3 devices found.

Do you want to continue? [confirm]

D1#

```
Jan  6 22:06:46.299: %SW_VLAN-4-VTP_PRIMARY_SERVER_CHG: d8bl.9028.af80 has
become the primary server for the VLAN VTP feature
```

DS-1# **show vtp status | i Primary**

DS-1#show vtp status | i Primary

```
VTP Operating Mode      : Primary Server
Primary ID              : 500a.0001.8000
Primary Description     : DS-1
```

Now we see that VTPv3 is working in that it allows for extended range VLANs while in Server mode, it is sending VTPv2-compatible messages at the domain boundary, and that DS-2 is still passing those messages along.

If you did not notice in the earlier output, verify the Configuration Revision number on DS-1, and then the number on AS-1. In this case, to cause a change at AS-1, DS-1 will have to have five revisions made to increment the revision number past what A1 currently has.

```
DS-1(config)# do show vtp status | i Revision
Configuration Revision : 1
```

```
A1# show vtp status | i Revision
Configuration Revision : 5
```

On DS-1, create the following VLANs.

- o VLAN 6, named SixthVLAN
- o VLAN 7, named SeventhVLAN
- o VLAN 8, named EighthVLAN
- o VLAN 9, named NinthVLAN
- o VLAN 10, named TenthVLAN
- o VLAN 11, named EleventhVLAN

Verify that they exist in the VLAN database and verify the configuration revision number is higher than the revision number AS-1 had in the previous step.

```
DS-1# show vlan brief | i Six|Seven|Eight|Nin|Ten|Elev
6      SixthVLAN          active
7      SeventhVLAN       active
8      EighthVLAN        active
9      NinthVLAN         active
10     TenthVLAN         active
11     EleventhVLAN      active
```

On DS-1, create VLAN 2600 with the name HackerPub. Verify that it exists in the VLAN database and verify the configuration revision number.

```
DS-1# config t
Enter configuration commands, one per line. End with CNTL/Z.
DS-1(config)# vlan 2600
DS-1(config-vlan)# name HackerPub
DS-1(config-vlan)# end

DS-1# show vlan brief | i Hack
2600 HackerPub          active
```

Step 2. Configure and observe VTPv3 domain operations.

In this step we will update DS-2 and AS-1 to run VTPv3 and verify their behavior as a Transparent and Client switches in the VTP domain.

Configure D2 and A1 to operate using VTP version 3.

```
DS-2# config t
Enter configuration commands, one per line. End with CNTL/Z.
DS-2(config)# vtp version 3
DS-2(config)# end

Jan  6 22:12:28.728: %SW_VLAN-6-OLD_CONFIG_FILE_READ: Old version 2
VLAN configuration file detected and read OK. Version 3
files will be written in the future.
```

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```
AS-1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
S-A1(config)# vtp version 3
AS-1(config)# end
Jan  6 22:12:50.784: %SW_VLAN-6-OLD_CONFIG_FILE_READ: Old version 2
VLAN configuration file detected and read OK.  Version 3
files will be written in the future.
```

Check the version and revision number on AS-1 and see that they have been updated to match DS-1. Verify that VLAN 2600 now exists in the VLAN database on AS-1.

```
AS-1# show vtp status | i
version|Domain|Revision|Primary|Mode|Feature
VTP version running           : 3
VTP Domain Name               : CCNPv8
VTP Pruning Mode              : Disabled
Feature VLAN:
VTP Operating Mode            : Client
Configuration Revision        : 8
Primary ID                    : d8b1.9028.af80
Primary Description            : D1
Feature MST:
VTP Operating Mode            : Transparent
Feature UNKNOWN:
VTP Operating Mode            : Transparent

AS-1# show vlan brief | i Hack
2600 HackerPub                active
```